MESA Red Meat Metabolites Project

LekkiWood (LekkiWood@Gmail.com)

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Important notes

- You should always check you have the most recent files! I use dynamic dates, so this can be done by checking that your files match the dates on the latest GitHub versions:
  - My profile can be found here
  - My repositories are listed here
  - The most recent version of this PDF README file can be found here
  - The most recent version of the source code for this README file can be found here
  - The most recent source code for creating the output files can be found here

Summary

Table 1: Duplicate IDs in raw amide assay data

sidno	exam	n
10509	5	2
14687	6	2
25200	5	2

Data Preparation (Clary - you can probably ignore this)

Metabolites

Input files

- A table of metabolite abundances from the amide assay: 25_0107_TOPMed_MESA_Amide-neg_rev031325.csv
- A table of metabolite abundances from the C8-pos assay: 24_1210_TOPMed_MESA_C8-pos_checksums_rev031325.csv
- A table of metabolite abundances from the C18-neg assay: 24_1210_TOPMed_MESA_C18-neg_checksums_rev031325.csv
- A table of metabolite abundances from the HILIC assay: 24_1210_TOPMed_MESA_HILIC-pos_checksums_rev031325.csv
- Sample information for the amide assay: MesaMetabolomics_PilotX01_AmideNeg_SampleInfo_20250329.txt
- Sample information for the C8-pos assay: MesaMetabolomics_PilotX01_C8Pos_SampleInfo_20250329.txt
- Sample information for the C18-neg assay: MesaMetabolomics_PilotX01_C18Neg_SampleInfo_20250329.txt
- Sample information for the HILIC assay: MesaMetabolomics_PilotX01_HILIC-Pos_SampleInfo_20250329.txt
- A file to bridge SHARe ids (sidno) with MESA IDs (idno) MESA-SHARE_IDList_Labeled.csv

Build Raw Metabolite Tables

Amide

- The amide sample info file contained information on N=13727 TOM IDs (each corresponding to one participant's abundances at one exam).
- Duplicates were identified (see Table 1) and removed.

Table 2: Duplicate IDs in raw C8 assay data

sidno	exam	n
10509	5	2
14687	6	2
25200	5	2

Table 3: Duplicate IDs in raw C8 assay data

sidno	exam	n
10509	5	2
14687	6	2

- Once the duplicates were removed, the final amide table contained data on N= 155 metabolites across 13721 observations.

C8

- The C8 sample info file contained information on N=13722 TOM IDs (each corresponding to one participant's abundances at one exam).
- Duplicates were identified (see Table 2) and removed.

- Once the duplicates were removed, the final amide table contained data on N= 869 metabolites across 13716 observations.

C18

- The C18 sample info file contained information on N=13684 TOM IDs (each corresponding to one participant's abundances at one exam).
- Duplicates were identified (see Table 3) and removed.

- Once the duplicates were removed, the final amide table contained data on N= 2655 metabolites across 13680 observations.

HILIC

- The HILIC sample info file contained information on N=13726 TOM IDs (each corresponding to one participant's abundances at one exam).
- Duplicates were identified (see Table 4) and removed.

Table 4: Duplicate IDs in raw HILIC assay data

sidno	exam	n
10509	5	2
14687	6	2
25200	5	2

- Once the duplicates were removed, the final amide table contained data on N= 906 metabolites across 13720 observations.
- The cleaned and transposed files from each assay were merged into a final, long form metabolite table with N=13726 observations on 4590 variables (5 variables are not metabolites: sidno, subject_id, TOM_ID, exam, idno).

Build Mapping file

- The following variables were created:
  - "Presence MESA X01" (coded 1)
  - "Presence MESA PILOT" (coded 1)
  - "Presence MESA MESA2" (coded 1).
- The original metabolite name (where known) from the lab was preserved as the variable 'Original_Metabolite_Name'
- A variable "Known_Compound" was created to indicate whether the compound had been assigned a name as of Wednesday, April, 22, 2026[2] (coding 1= yes, 0=no).
- For unknown compounds, the lab ID (arbitrarily assigned by The Broad) assigned within the X01 study was used, and prefixed with the assay name.
- The number of known/unknown compounds per assay is:
  - Amide: N= NULL known compounds and no unknown compounds.
  - C18: N= NULL known compounds and NULL unknown compounds.
  - C8: N= NULL known compounds and NULL unknown compounds.
  - HILIC: N= NULL known compounds and NULL unknown compounds.
- The final mapping file has information on NULL known compounds and NULL unknown compounds

Calculate QC data (missingness and CV)

Coefficients of Variation (CV)

Table 5: Coefficients of Variation by assay

Assay	Mean_CV
Amide	11.48314
C18	14.49480
C8	11.50480
HILIC	14.04263

Table 6: Coefficients of Variation by assay and exam

Known	Mean_CV
Amide	
Known Metabolites	11.48
C18	
Unknown Compounds	14.90
Known Metabolites	6.61
C8	
Unknown Compounds	15.60
Known Metabolites	5.37
HILIC	
Known Metabolites	14.04

- For each of the four metabolite tables above, the QC samples for calculating the coefficient of variation (CV) were selected by selecting Sample_Type=="QC-pooled_ref" from the sample info files.
- CVs were calculated across all exams (since batches were not stratified by exam) as the mean of all QC samples, divided by the standard deviation of all QC samples, multiplied by 100, and saved as the variable "cv_percent".
- Across, all assays, the mean CV was 13.7398974, with a range of 1.4705788 - 666.0002291.
- The CVs per assay, in Table 5:

- The CVs per assay, stratified into known and unknown metabolites are in Table 6:

Missingness

- The proportion of missingness for each assay, by exam is in Table 7 (note, this is for people with assay data at each exam, missingness does not include, for example, people

Table 7: Missingness by assay, and by exam

Assay	Overall	E1	E5	E6
Amide	18.831274	19.2825806	22.044571	13.2412389
C18	6.181398	7.1368650	5.248741	5.5028499
C8	0.893234	0.7809743	1.017541	0.9517183
HILIC	9.420818	10.2592391	7.988617	9.7106041

who have data at exam 1, but have no data at exam 5):

Removing duplicate metabolites

Annotation cleaning

- Some light cleaning of the metabolite annotations was conducted (note: this is an ongoing process):
- Glutamic acid had different HMDBs on the original HILIC and amide assays (see Table 8). Per advice: The HMDB from the HILIC was retained for both assays, and these were treated as the same compound.

Table 8: HMDBs for Glutamic Acid in raw data

Lab ID	HMDB	Metabolite	Assay
QI18171	HMDB0000148	Glutamic acid	hilic
Glutamic.acid	HMDB0003339	Glutamic acid	amide

- The metabolite Glutamic acid had the HMDB ID changed from HMDB0003339 `->` HMDB0000148.
- For the amide assay, the compound with HMDB ID "HMDB0000122" had a different name to that used for the HILIC and C18 assays (see Table 9). Both names atr "correct", just different conventions, so these were treated as the same compounds when screening for duplicates.

Table 9: Annotations for HMDB0000122 in raw data

Lab ID	HMDB	Metabolite	Assay
Glucose.Fructose. Galactose_waterloss	HMDB0000122	Glucose/Fructose/ Galactose_waterloss	amide
QI3656	HMDB0000122	Hexose	c18
QI14125	HMDB0000122	Hexose	hilic

Table 10: Information on metabolites included on more than one assay

Metabolite Duplicate status	Frequency (N)
Unique or missing HMDB ID	4343
Duplicated HMDB ID with lowest CV	116
Internal standard	5
Duplicated HMDB_ID and not lowest CV	121

Removing duplicated metabolites

- The HMDB_ID variable was used to flag compounds measured on more than one assay
- To flag metabolites included on more than one asssay, a variable was created "Retain", with the following coding:
  - 0, labelled "Unique or missing HMDB ID": an unknown compound, or a unique HMDB ID
  - 1, labelled "Duplicated HMDB ID with lowest CV": A duplicated HMDB ID with the lowest coefficient of variation (CV) across all compounds sharing that HMDB ID
  - 2, labelled "Internal standard": A compound used as an internal standard (generally excluded from standard analyses)
  - 3, labelled "Duplicated HMDB_ID and not lowest CV": A duplicated HMDB ID that does *not* have the lowest coefficient of variation (CV) across all compounds sharing that HMDB ID (generally excluded from analyses where you do not want the same compound included more than once)
- See Table 10
- All metabolites that had either a unique HMDB ID, or that had a duplicated HMDB ID but had the lowest CV from all those metabolites with the same HMDB ID were retained, all others were removed from the analysis.
- The final metabolite file contained information on N=4458 metabolites across N=13726 observations.

Some unresolved issues to note

- The raw data from the C18 assay has a smaller N than some others
- Most duplicated compounds are measured on different assays. One exception is LPC 16:0/0:0, which is measured twice on the C8 assay, with the same HMDB, but different assigned LAB IDs. For now, the compound with the Compound ID in the X01 of "TF04" is renamed as LPC 16:0/0:0_v2.

- Some lipids have the same HMDB ID, but the different common metabolite names, and these names are not listed as synonyms in the HMDB (e.g., Cer 40:1;O2 (C8 assay) is not listed as a synonym for 18:1 (HILIC), and Cer 42:2;O2_A (C8) is not listed for a synonym for Cer 18:1;O2/24:1 (HILIC). For now, these are treated as separate compounds (see Table 11 and Table 12).

Table 11: Annotations for HMDB0004952 in raw data

Lab ID	HMDB	Metabolite	Assay
QI04610	HMDB0004952	Cer 18:1;O2/22:0	hilic
QI8023	HMDB0004952	Cer 40:1;O2	c8

Table 12: Annotations for HMDB0004952 in raw data

Lab ID	HMDB	Metabolite	Assay
QI04540	HMDB0004953	Cer 18:1;O2/24:1	hilic
QI4265	HMDB0004953	Cer 42:2;O2_A	c8

Methods

Procedure

Analyses

PCA

Data selection

As the modeling approach cannot accommodate missing data, a data-adaptive feature selection procedure to balance predictor inclusion with retention of participant-level completeness. Specifically, a greedy, forward-selection algorithm was applied in which candidate features (metabolites) were evaluated in blocks of 50 to improve computational efficiency. Within each iteration, the feature whose inclusion resulted in the smallest reduction in the number of eligible participants—defined as those with non-missing outcome data for true class membership—was selected from the evaluated block. Following selection, participant eligibility was recalculated based on complete-case availability for all included features. This process was repeated iteratively across successive blocks of candidate features, with features prioritized according to their minimal impact on sample retention. The procedure continued until inclusion of an additional feature would have reduced the analytic sample below a prespecified threshold of two-thirds of eligible participants.

Unsupervised analysis of underlying structure

To explore the underlying structure of the metabolite data without a priori hypotheses, an unsupervised principal component analysis (PCA) was conducted using the mixOmics package in R. PCA was applied to the set of selected features to reduce dimensionality and identify dominant axes of variation across participants. Prior to PCA analysis, continuous variables were standardized to have mean zero and unit variance to ensure comparability across features. The PCA algorithm decomposes the data into orthogonal components that successively capture the greatest variance in the dataset, with the first few components summarizing the primary patterns of variability. Component scores were used to visualize clustering and heterogeneity among participants, and loading vectors were examined to identify features contributing most strongly to each principal component. This unsupervised approach provided a data-driven summary of the multivariate structure, facilitating interpretation of complex relationships among features.

Because this is an exploratory analysis for data QC, and not for prediction, we did not use training and test datasets - the PCA was just run on the whole sample after selection via the greedy algorithm.

Results

Participant characteristics

Clary - tell me if you need this.

PCA

- The greedy algorithm selected N=`length(included_vars$Metabolite)` metabolites to include.
  - A list of these can be found here.
  - A total of N=51.
  - A total of N=581 were from the HILIC assay.
  - A total of N=727 were from the C8 assay
  - A total of N=1545 were from the C18 assay
- A total of N=4780 MESA participants from exam 1 were included.

A scree plot suggested that a single component was chosen for the PCA Figure 1.

A PCA was fit to the metabolite data with a single component.

- Individual component scores with various MESA IDs / TOPMed IDs can be found here.
- Variable loadings for each metabolite can be found here.

There was clear clustering of the MESA participants within component 1 Figure 2. This did not seem to be due to gender Figure 3 (I also checked age, BMI, diabetes status, and lipids, as well as the metabolite technical covariates I could think of).

Assay specific results.

The loadings of the metabolites onto component 1, by assay, are in Figure 4.

PCAs were run for each assay seperately, and the distribution of the extracted component plotted:

- C8: Figure 5
- C18: Figure 6
- Amide: Figure 7

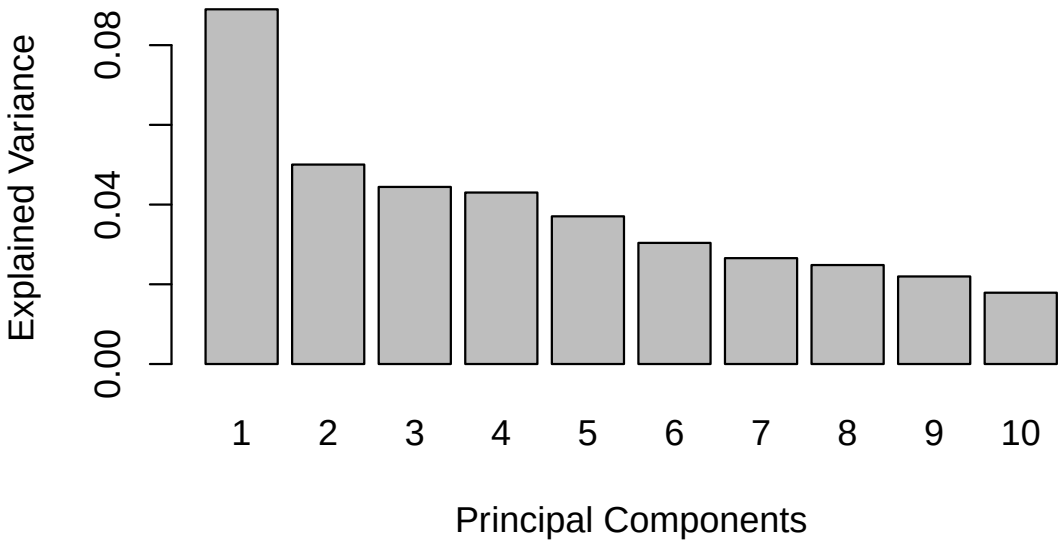

Figure 1: Screeplot from the PCA performed on the MESA metabolite data

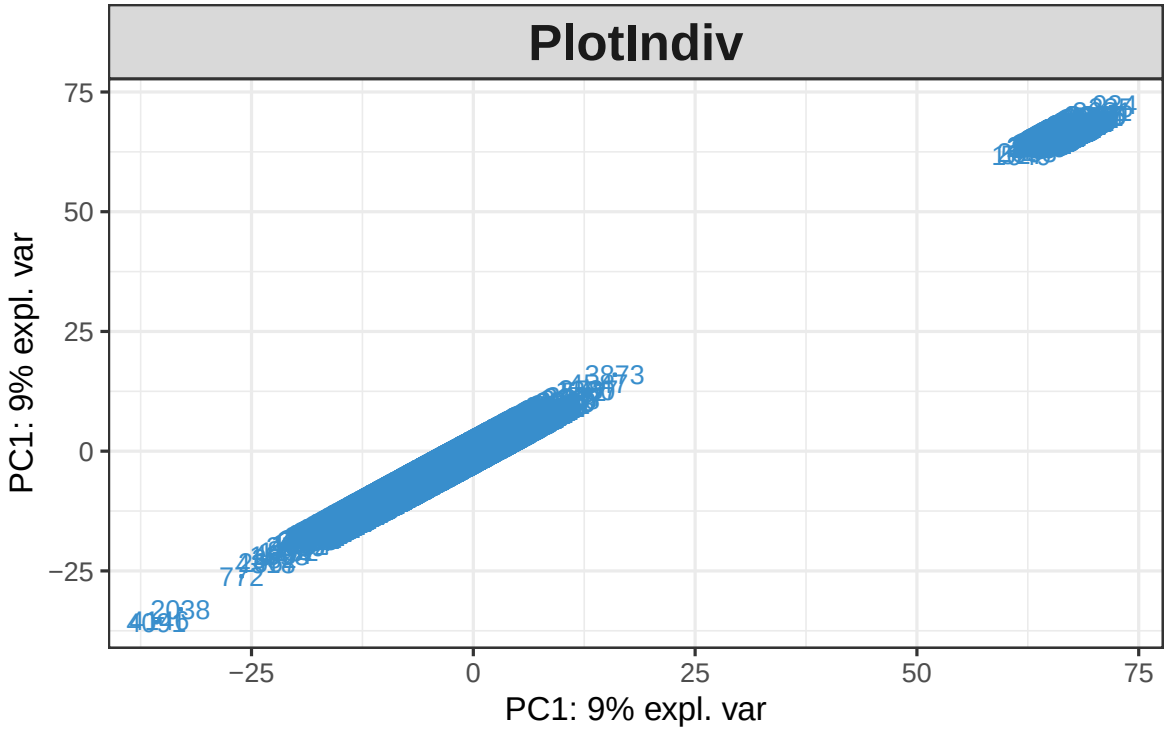

Figure 2: Distribution of scores for component 1

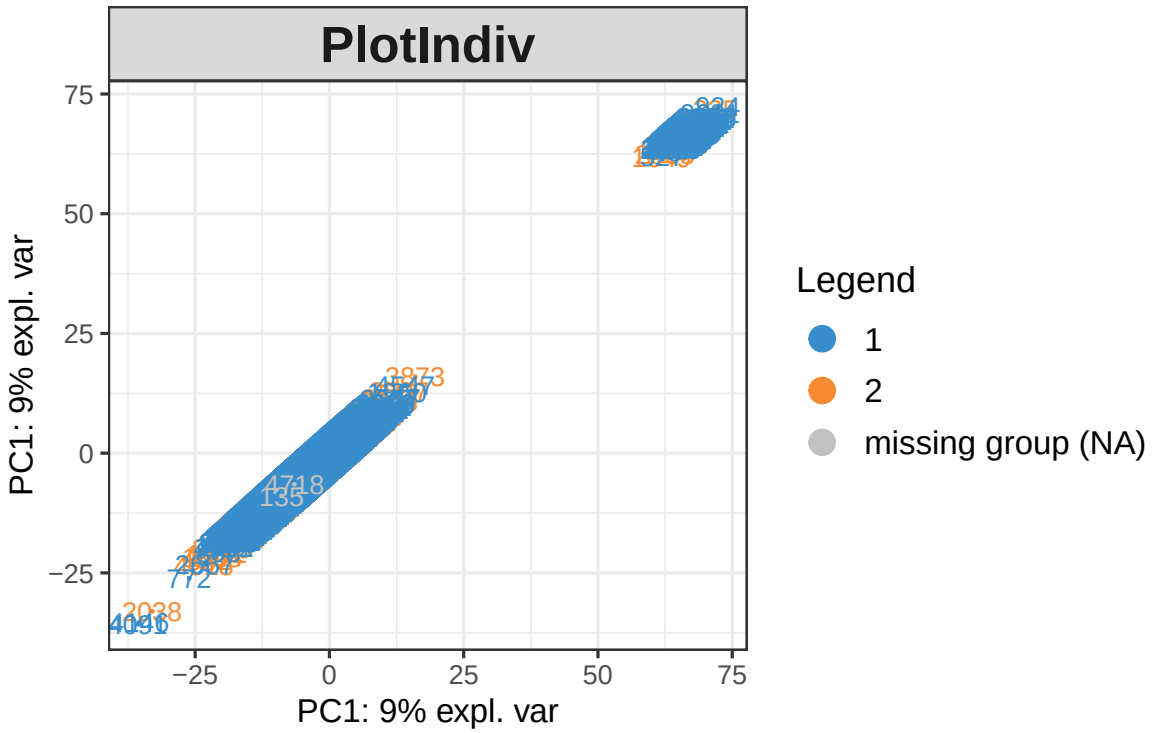

Figure 3: Distribution of scores for component 1 by gender

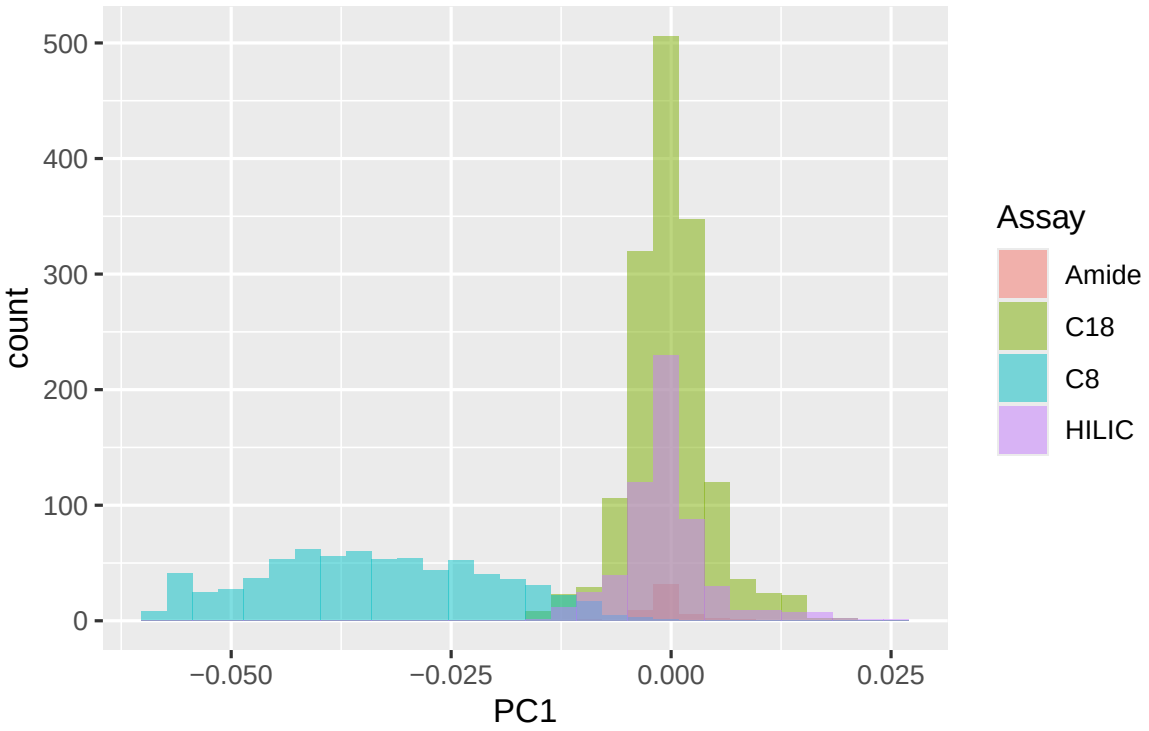

Figure 4: Distribution of scores for component 1 by gender

- HILIC: Figure 8

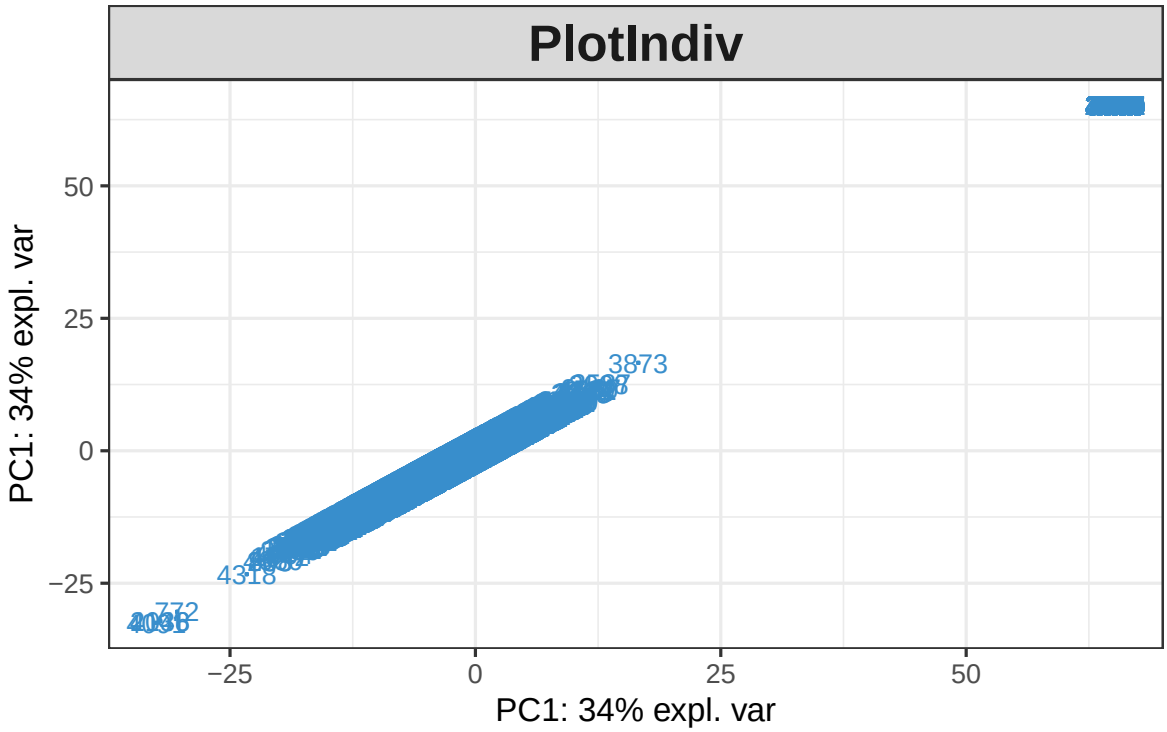

Figure 5: Distribution of scores for component 1 for C8

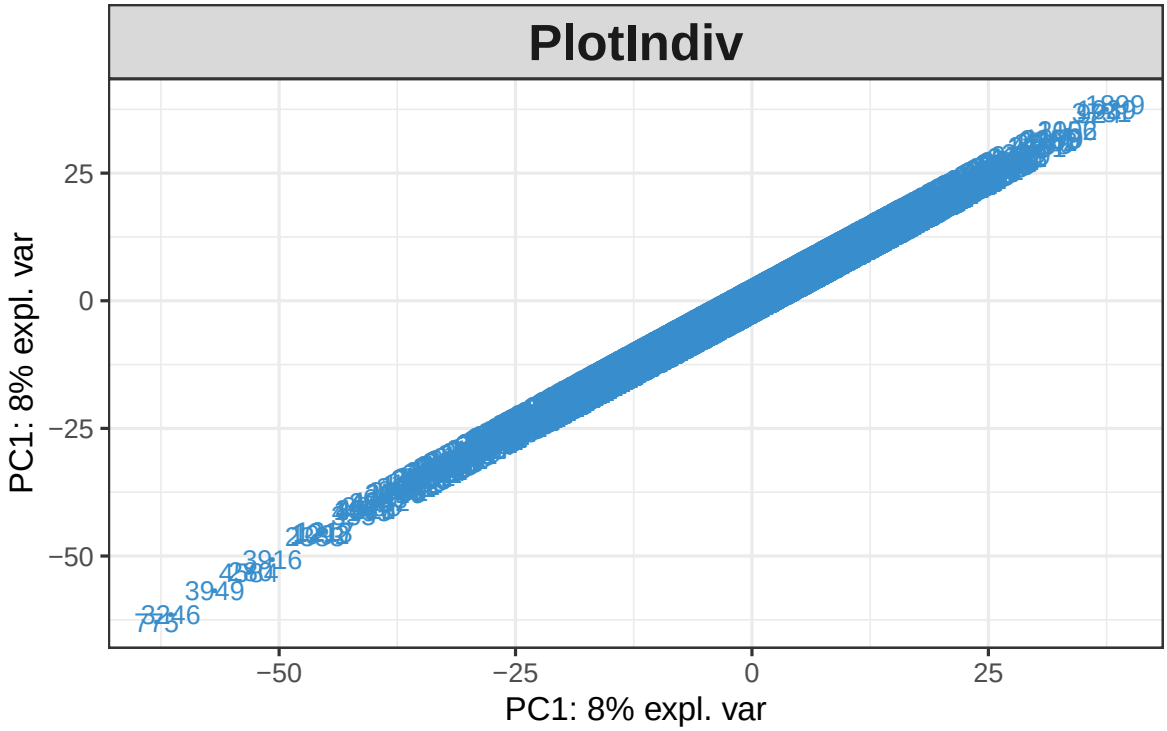

Figure 6: Distribution of scores for component 1 for C18

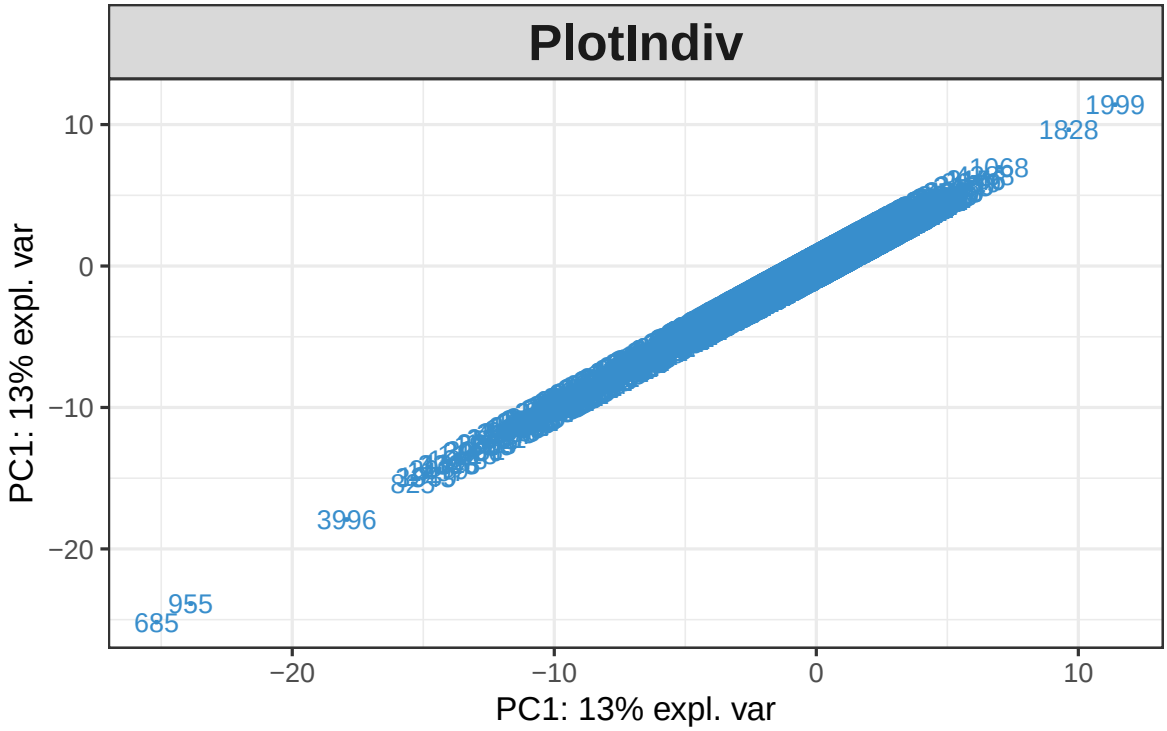

Figure 7: Distribution of scores for component 1 for Amide

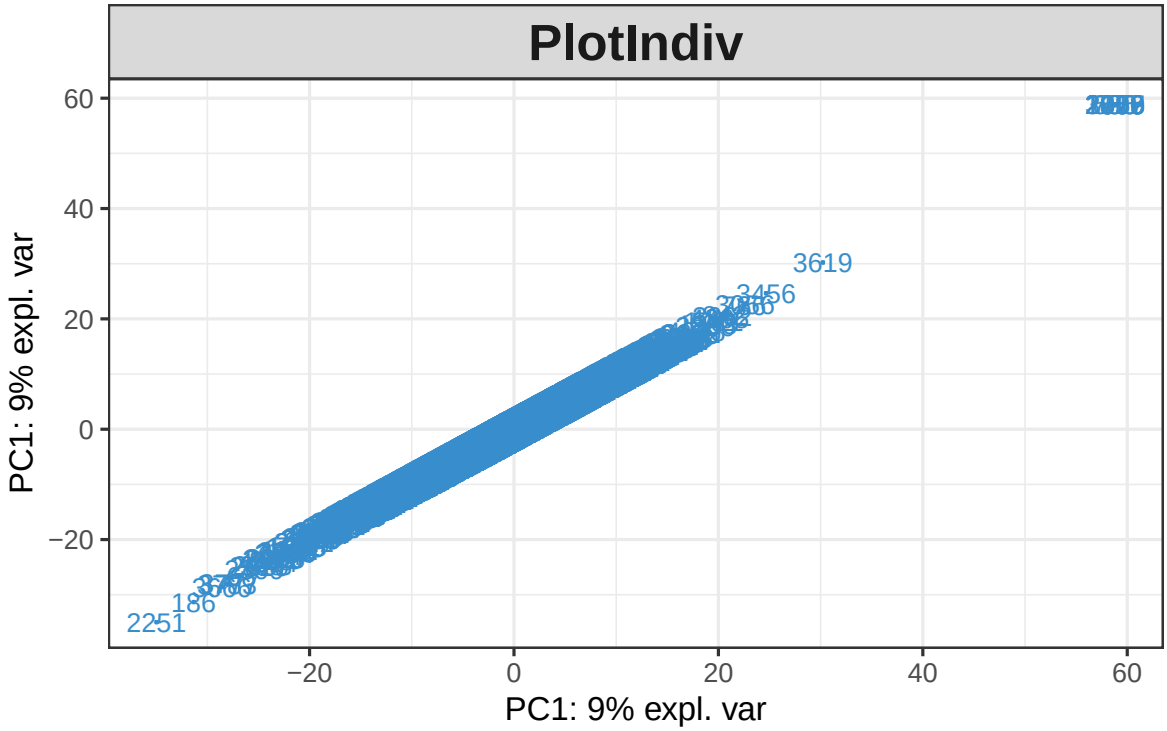

Figure 8: Distribution of scores for component 1 for HILIC